Atomic Structure and the Periodic Table

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2.1 : ATOMIC STRUCTURE

Specification point 2.1

Nuclear Structure

An atom consists of a central, extremely small, dense nucleus containing protons and neutrons (collec-

> tively termed nucleons) 4, 5.

Extranuclear Region

> Electrons orbit around the nucleus in defined quantum energy levels / shells 4, 5.

Atomic Volume

> Most of an atom's volume is empty space surrounding the dense nucleus 4.

2.2 : RELATIVE MASS AND CHARGE

Specification point 2.2

Subatomic Particle,Relative Mass,Relative Charge,Location in Atom

SUBATOMIC PARTICLE	RELATIVE MASS	RELATIVE CHARGE	LOCATION IN ATOM
Proton	1	+1	Nucleus 1, 5, 6
Neutron	1	0 (Neutral)	Nucleus 1, 5, 6
Electron	$\frac{1}{1836}$ (or 0.00055)	-1	Quantum shells 1, 4-6

2.3 : ATOMIC NUMBER AND MASS NUMBER

Specification point 2.3

Atomic (Proton) Number (Z)

> The total number of protons in the nucleus of an atom of an element 1, 6.
>
> It uniquely defines the chemical identity of the element.

Mass Number (A)

> The total number of protons plus neutrons (nucleons) in the nucleus of an atom 1, 6.

2.4 : SUBATOMIC PARTICLE CALCULATIONS

Specification point 2.4 ▾

Particle count from Z and A			
	Neutral Atom	**Cation (X^{n+})**	**Anion (Y^{m-})**
CHARGE CHANGE	No electrons gained or lost.	Formed when electrons are lost.	Formed when electrons are gained.
PROTONS	Protons = Z	Protons = Z	Protons = Z
ELECTRONS	Electrons = Z	Electrons = Z - n	Electrons = Z + m
NEUTRONS	Neutrons = A - Z	Neutrons = A - Z	Neutrons = A - Z

2.5 : ISOTOPES

Specification point 2.5

2.6 : MASS SPECTROMETRY

Specification point 2.6

BASIC OPERATING STEPS

1 Vapourisation

> Sample is converted to gas 12.

2 Ionisation

> High-energy electrons knock out electrons from sample atoms to produce positive ions: $\mathrm{X(g) + e^- \rightarrow X^+(g) + 2e^-}$ 1, 12.

3 Acceleration

> Positive ions are accelerated by an electric field to give equal kinetic energy 12.

4 Deflection

> Magnetic field deflects ions based on their mass-to-charge ratio (m/z) 12.
>
> Lighter or more highly charged ions experience greater deflection 12.

5 **Detection**

> Ions hit a detector generating an electric current proportional to abundance 12.

2.7 : DIATOMIC MASS SPECTRA

Specification point 2.7

Chlorine Gas (Cl_2) Mass Spectrum

- Chlorine naturally consists of ^{35}Cl (75% / $\frac{3}{4}$) and ^{37}Cl (25% / $\frac{1}{4}$) 11, 17.

Chlorine molecular-ion peaks

M/Z	ISOTOPE COMBINATION	PROBABILITY
70	$^{35}\mathrm{Cl}-^{35}\mathrm{Cl}^{+}$	$\frac{3}{4}\times\frac{3}{4}=\frac{9}{16}$ 17.
72	$^{35}\mathrm{Cl}-^{37}\mathrm{Cl}^{+}$ / $^{37}\mathrm{Cl}-^{35}\mathrm{Cl}^{+}$	$2\times(\frac{3}{4}\times\frac{1}{4})=\frac{6}{16}$ 17.
74	$^{37}\mathrm{Cl}-^{37}\mathrm{Cl}^{+}$	$\frac{1}{4}\times\frac{1}{4}=\frac{1}{16}$ 17.

PEAK HEIGHT RATIO

$$m/z = 70 : 72 : 74 = 9 : 6 : 1$$

17.

Bromine Gas (Br_2) Mass Spectrum

Bromine molecular-ion peaks

ISOTOPES	M/Z	PEAK RATIO
$^{79}\mathrm{Br}$ **(50%) and** $^{81}\mathrm{Br}$ **(50%) 18.**	158, 160, 162	1 : 2 : 1 18.

2.8 : IONISATION ENERGIES

Specification point 2.8

First Ionisation Energy (IE_1)

> The energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous 1+ ions 11, 19-21.
>
> $$\mathrm{X(g)} \rightarrow \mathrm{X^{+}(g)} + \mathrm{e^{-}}$$

Second Ionisation Energy (IE_2)

> $\mathrm{X^{+}(g)} \rightarrow \mathrm{X^{2+}(g)} + \mathrm{e^{-}}$ 11, 22, 23

Third Ionisation Energy (IE_3)

$X^{2+}(g) \rightarrow X^{3+}(g) + e^-$ 11

2.9 : ORBITALS AND SUBSHELL CAPACITY

Specification point 2.9

s-subshell

1 orbital → max 2 electrons 24, 26.

p-subshell

3 orbitals → max 6 electrons 24, 26.

d-subshell

5 orbitals → max 10 electrons 24, 26.

---

f-subshell

7 orbitals → max 14 electrons 24, 26.

2.10 : FACTORS AFFECTING IONISATION ENERGY

Specification point 2.10

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1. Nuclear Charge (Protons)

> More protons → stronger electrostatic pull on outer electrons → higher IE 20, 24, 27, 28.

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2. Atomic Radius / Distance

> Outer electron further from nucleus → weaker electrostatic attraction → lower IE 24, 27.

3. Electron Shielding

> Inner shell electrons repel outer electrons, shielding them from full nuclear attraction $\rightarrow$ lower IE 24, 27, 29.

4. Subshell Energy & Pairing Repulsion

> Electrons in higher energy subshells (3p vs 3s) or paired orbitals (electron-electron repulsion) require less energy to remove 24, 29-31.

2.11 : EVIDENCE FOR QUANTUM SHELLS AND SUBSHELLS

Specification point 2.11 ▾

How ionisation-energy data reveals electronic structure

	Quantum Shell Evidence	Sub-shell Evidence
EVIDENCE	Large jumps between successive ionisation energies occur when an electron is removed from a new quantum shell closer to the nucleus (experiencing drastically less shielding) 24, 32-34.	Small dips in first ionisation energy across Period 2 (Be → B and N → O) and Period 3 (Mg → Al and P → S) prove that main quantum shells are subdivided into s and p sub-shells 24, 29-31.

WHAT IT TELLS YOU	The number of electrons removed before the first huge jump equals the number of outer shell electrons, revealing the element's Group number 32-34.	First ionisation energies of successive elements provide evidence for electron sub-shells.

2.12 : SHAPES OF S AND P ORBITALS

Specification point 2.12

s-orbital

Spherical / ball-shaped 24, 35.

p-orbital

Dumbbell-shaped (aligned along x, y, or z axes) 7, 24, 35.

2.13 : ELECTRON FILLING IN ORBITALS

Specification point 2.13

Two rules for orbital filling		
	Hund's Rule	**Pauli Exclusion Principle**
SOURCE NOTE	Degenerate orbitals (orbitals of equal energy, e.g. $2p_x$, $2p_y$, 2p_z) fill singly with parallel spins before pairing occurs 26.	An orbital can accommodate a maximum of two electrons, which must possess opposite spins ($\uparrow\downarrow$) to minimise mutual repulsion 26.

2.14 : ELECTRONIC CONFIGURATION

Specification point 2.14

FILLING SEQUENCE

$$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d$$

26.

EXCEPTION

Chromium (Z = 24) $[Ar]\,3d^5 4s^1$

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$ (half-filled 3d subshell provides extra stability) 3, 26.

EXCEPTION

Copper (Z = 29) $[Ar]\,3d^{10}4s^{1}$

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$ (fully filled 3d subshell provides extra stability) 26, 36.

2.15 : ELECTRONIC CONFIGURATION AND CHEMICAL PROPERTIES

Specification point 2.15

Outer shell (valence) electron configuration

> Outer shell (valence) electron configuration dictates chemical reactivity, bonding behaviour, and valency 26, 37.

Same group

> Elements in the same group have identical outer shell configurations and therefore undergo similar chemical reactions 26, 37.

Isotopes

> Isotopes have identical electronic configurations, giving them identical chemical properties 37, 38.

2.16 : PERIODIC TABLE BLOCKS AND SHELL CAPACITIES

Specification point 2.16

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s-block

Groups 1 & 2 (outermost electron in an s-orbital) 26, 39.

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p-block

Groups 3 to 8/0 (outermost electron in a p-orbital) 26, 40.

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d-block

Transition metals (filling d-orbitals) 26, 41.

Quantum Shell Capacities (\(n=1\text{ to }4\))		
QUANTUM SHELL	**CONFIGURATION**	**CAPACITY**
n=1	$1s^2$	2 electrons
n=2	$2s^2\ 2p^6$	8 electrons
n=3	$3s^2\ 3p^6\ 3d^{10}$	18 electrons
n=4	$4s^2\ 4p^6\ 4d^{10}\ 4f^{14}$	32 electrons 26.

2.17 : PERIODIC PROPERTIES AND LOGARITHMIC GRAPHS

Specification point 2.17

Logarithmic Graphs ($\log_{10} IE$)

> Because successive ionisation energies span huge numerical ranges (from hundreds to tens of thousands of $kJmol^{-1}$), plotting $\log_{10}(\text{Ionisation Energy})$ vs electron removed turns exponential jumps into distinct linear steps, allowing clear identification of quantum shell boundaries 40, 42, 46.

2.18 : PERIOD 2 AND 3 TRENDS

Specification point 2.18

i. Melting and Boiling Temperature Trends

Metallic Elements ($\text{Li, Be, Na, Mg, Al}$)

> Melting points increase from Group 1 to Group 3 because ionic charge increases, more delocalised

> electrons are donated per cation, and ionic radii decrease → stronger metallic bonding 47, 48.

Giant Covalent Elements ($\mathrm{B}, \mathrm{C}, \mathrm{Si}$

> Highest melting temperatures in Periods 2 & 3 42, 45, 49.
>
> Diamond, graphite, and silicon possess giant 3D macromolecular structures with strong covalent bonds extending throughout the lattice, requiring vast thermal energy to break 42, 50.

Simple Molecular Elements ($\mathrm{P_4}, \mathrm{S_8}, \mathrm{Cl_2}$) & Noble Gases ($\mathrm{Ar}$)

> Low melting points due to weak London dispersion forces between discrete molecules 42, 45, 47, 51.

MELTING POINT ORDER

$$S_8 > P_4 > Cl_2 > Ar$$

\(\text{S}_8\) has the highest melting point among the simple molecules because it is a larger molecule with more electrons, creating stronger London dispersion forces 42, 51, 52.

ii. First Ionisation Energy Trends Across Periods 2 & 3

General Trend

> IE_1 increases across Period 2 and Period 3 20, 42, 44, 45, 53.

Explanation

> Proton number / nuclear charge increases, atomic radius decreases, and shielding remains relatively constant → stronger attraction on outer electrons 20, 27, 28, 54.

Drop 1: Group 2 to Group 3 ($\mathbf{Be \rightarrow B}$ and $\mathbf{Mg \rightarrow Al}$)

> The outer electron in B/Al is in a p-subshell ($2\mathrm{p}$ or $3\mathrm{p}$), which is at a higher energy level and shielded by the inner s-electrons ($2\mathrm{s}$ or $3\mathrm{s}$) → requires less energy to remove 29, 42, 48, 55.

Drop 2: Group 5 to Group 6 ($\mathbf{N \rightarrow O}$ and $\mathbf{P \rightarrow S}$)

> In O/S, the electron is removed from a paired p-orbital 30, 31, 42.
>
> Inter-electron repulsion between the two paired electrons in the same orbital makes it easier to remove 30, 31.

iii. Decrease in First Ionisation Energy Down a Group

2.X : ADDITIONAL CONTEXT — NOT PART OF THE SOURCE NOTES

Ionisation energy mental model

Treat first ionisation energy as a competition between nuclear attraction and the factors that weaken it. When explaining a trend, identify nuclear charge, distance, shielding, and—where relevant—the subshell or electron-pairing difference.

Mass-spectrum shortcut

For a diatomic molecule, list every possible isotope combination, calculate the m/z value, then calculate each relative probability. Those relative probabilities give the relative peak heights.

Three layers of electronic structure

Keep shell, subshell and orbital separate: a shell is a principal energy level; a subshell is a division within a shell; an orbital is a region within a sub-shell that holds up to two electrons.

Trend-answer checklist

Do not stop at 'nuclear charge increases'. Name the changes in nuclear charge, atomic radius and shielding, then connect them to electrostatic attraction and the energy required.

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